Class-Aware Multifaceted Approach for Imbalanced Dermoscopic Image Classification

Master's Thesis Proposal

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Abstract and Introduction

Introduction : The Problem

Class Imbalance is the critical issue in skin cancer classification, especially for **Melanoma (mel).**

Why it matters

Melanoma is a rare but deadly form of skin cancer, and early detection is crucial for survival. Models often **fail to detect the minority class (low recall).**

Our Goal

To create a method that helps the model learn better on minority classes, enhancing its **reliability for diagnostic support** in clinical settings.

Approach

Hybrid method combining class-specific data augmentation, Focal Loss, and CBAM attention into one model.

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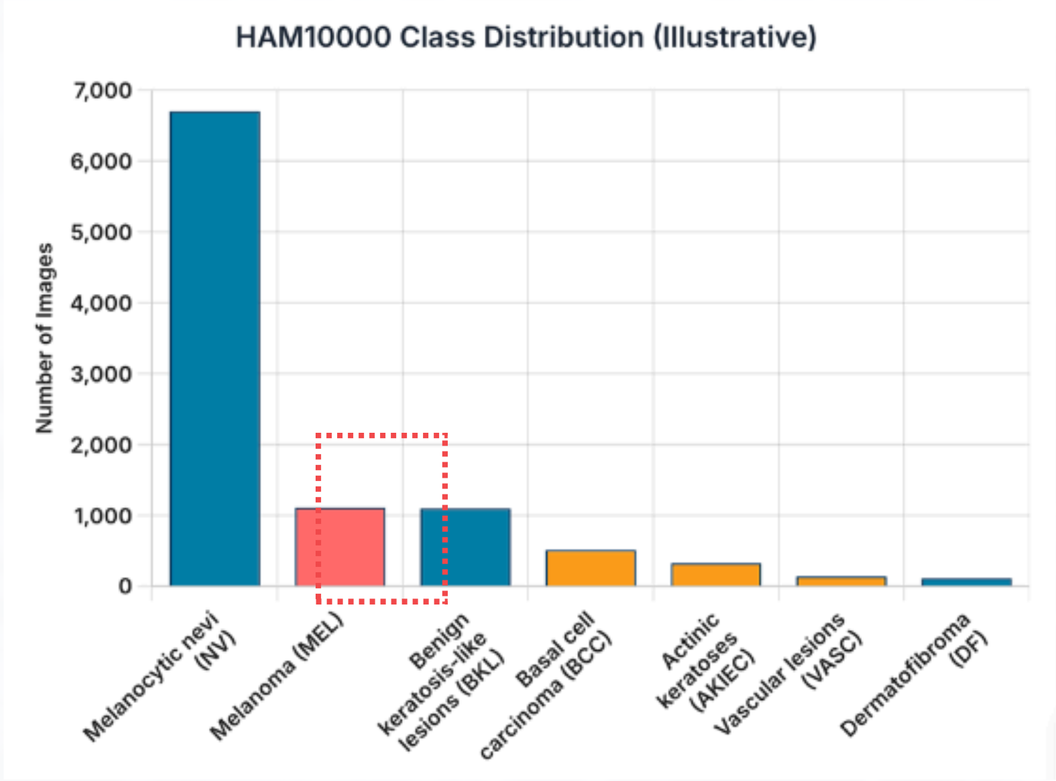
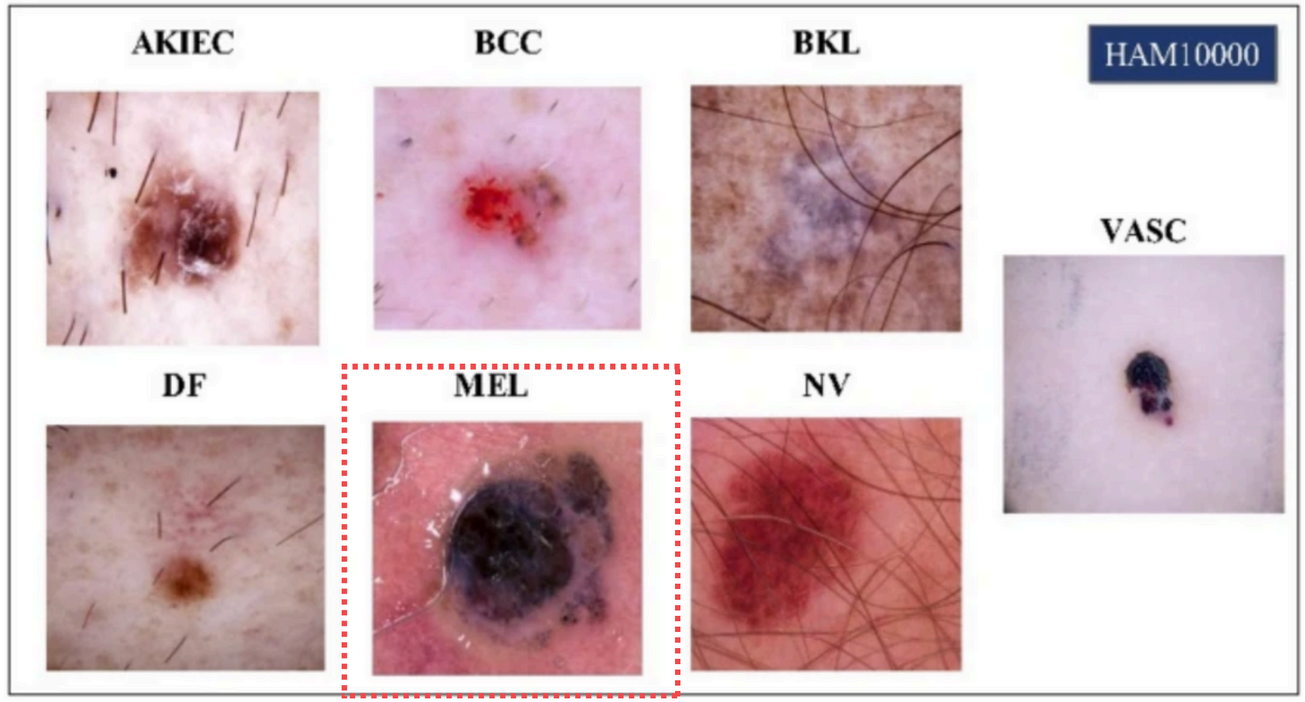


https://www.mdpi.com/2075-4418/15/1/99



Sample images of ISIC2018 and HAM10000 dataset

https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-025-03051-2

Problem Statement & Research Gap

Core Problem

*The **HAM10000** dataset contains over **67% benign cases**, causing deep learning models to perform poorly on critical minority classes like Melanoma.*

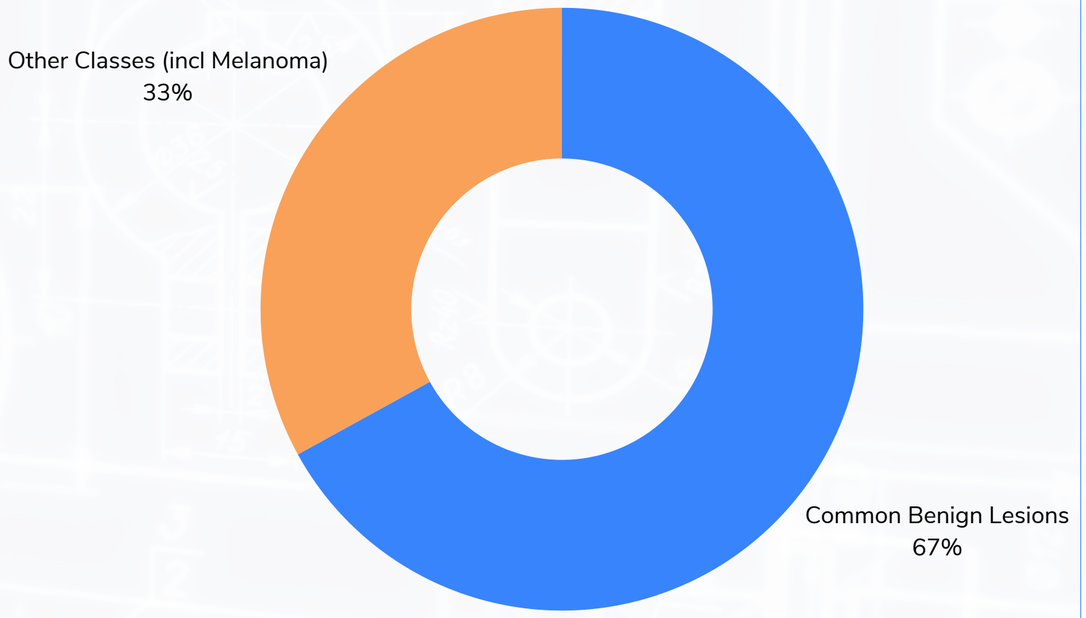
Limitation of Existing Works

- *Current systems often rely on **generic oversampling** or simple augmentation.*
- *They **fail to address the unique visual characteristics** of rare lesions.*
- *Standard loss functions do not sufficiently reduce misclassifications of **hard-to-diagnose lesions.***

Our Contribution

*We introduce a **Class-Aware Multifaceted Approach** to significantly enhance minority-class detection, resulting in a more robust and performance-enhanced model.*

Majority vs. Minority



Research Objectives

01

Core

To improve melanoma and other minority-class detection by using a ***class-aware multifaceted approach.***

02

Data Level

*To apply c****lass-specific data augmentation*** *and* ***Focal Loss*** *to handle the strong class imbalance.*

03

Model Level

To build a new model that combines **ResNet50** *and the* ***CBAM attention*** *algorithm to help the model focus more on important lesion areas.*

04

Evaluation

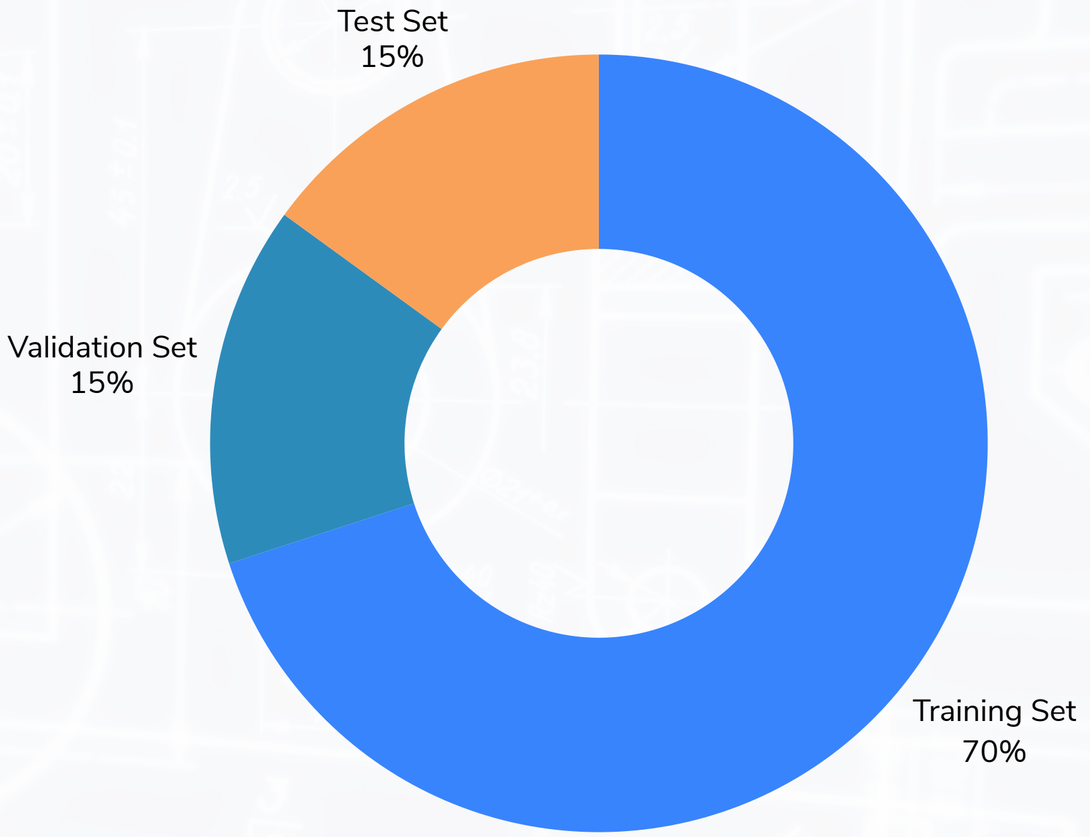
*Use R****ecall (Sensitivity), Macro-F1, and Balanced Accuracy*** *to ensure the results are good for minority classes.*

Literature Review

	Alsaidi (2023)	Cui (2025)	**Ourwork**
Scope	*Binary*	*Localization*	***Multi - Class***
Data	*Augmentation*	*GAN*	***Hybrid + Adaptive***
Algorithm	*Focal Loss*	*CBAM*	***Full Hybrid***
Integration	*Data + Loss*	*Attention + Loss*	***Multi - Class***
Outcomes	↑ *Recall*	↑ *Sensitivity*	↑ ***Balance + Explainability***

Metgodology : Data & Preprocessing

- **Data Source:** ***HAM10000*** *dataset (10,015 dermoscopic images with 7 skin classes).*
- **Preprocessing:** *Resizing and Normalization using ImageNet statistics.*
- **Data Partitioning (Crucial):** ***Using Stratified Sampling to ensure every class has the same ratio in all splits (70% Train, 15% Validation, 15% Test).***
- **Tools:** *Python, PyTorch,* ***Albumentations*** *(for augmentation), and Grad-CAM (for interpretability).*



Metgodology : Our 3-in-1 Strategy

1. Data-Level

Class-Specific Augmentation

Augment images for Minority Classes (Melanoma) using Geometric and Photometric Adjustments.

2. Algorithm-Level

Focal Loss

Makes the model focus on hard-to-classify examples over easy-to-classify majority examples.

3. Model Enhancement

CBAM Attention

Helps the model achieve a "Smart Focus" by emphasizing the Lesion Area and ignoring the Background.

Methodology : Model Architecture

Main Model (Our Proposal)

ResNet50 + CBAM

- ***ResNet50*** *is chosen for its reliability in medical image tasks.*
- ***CBAM*** *is integrated to boost attention to small, critical melanoma features.*

Baseline Model (Control)

EfficientNet-B0

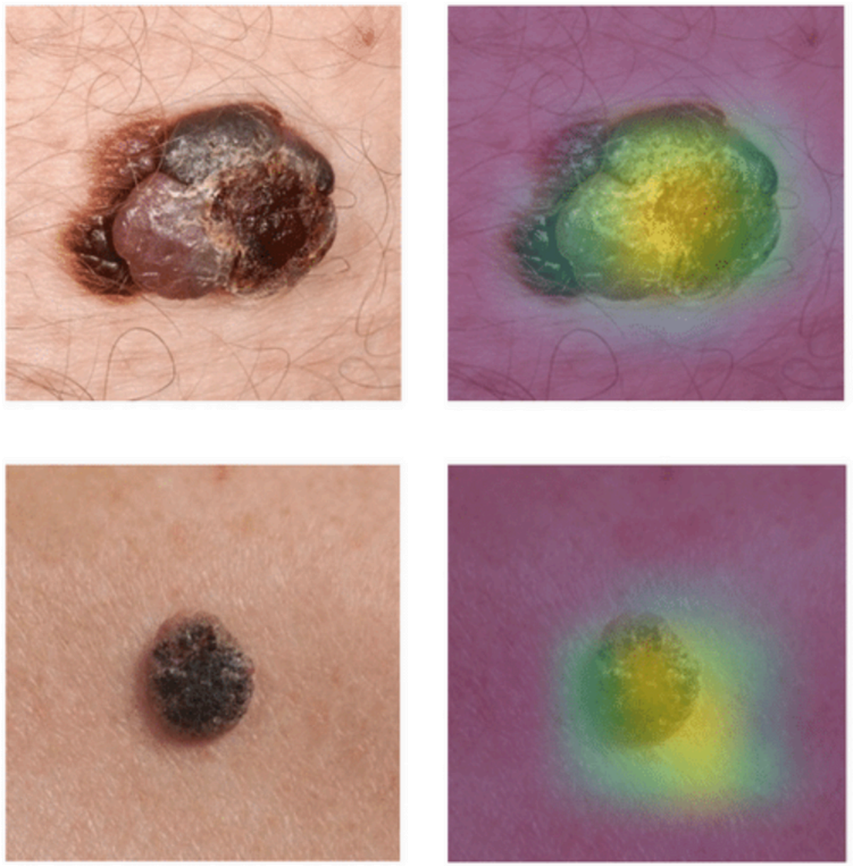
- *Used for a convincing comparison against a modern, effective architecture.*
- *Proves the merit of our multifaceted approach.*

Training: User Transfer Learning with ImageNet pre-trained weights

Evaluation Strategy

Why Overall Accuracy is Misleading

Standard accuracy is unreliable for imbalanced data.



Primary Metrics (Focus on Minority Class)

- ***Melanoma Recall (Sensitivity):*** *Crucial measure of the model's ability to correctly find the deadly class.*
- ***Macro-Averaged F1-Score:*** *Measures overall model balance across all 7 classes.*
- ***Balanced Accuracy:*** *A fairer metric for imbalanced datasets.*

Validation & Interpretability

- ***Validation:*** *Use Stratified k-fold cross-validation and Early Stopping to prevent overfitting.*
- ***Interpretability:*** *Grad-CAM heatmaps are used to visually confirm the model is focusing on the lesion features.*

Expected Outcomes

Primary Deliverable

A ResNet50 + CBAM model with high scores for ***Macro-F1 and Melanoma Recall.***

Key Findings

Expected to show that the combined, ***Class-Aware Multifaceted Method works*** *better than a normal model like EfficientNet-B0.*

Impact

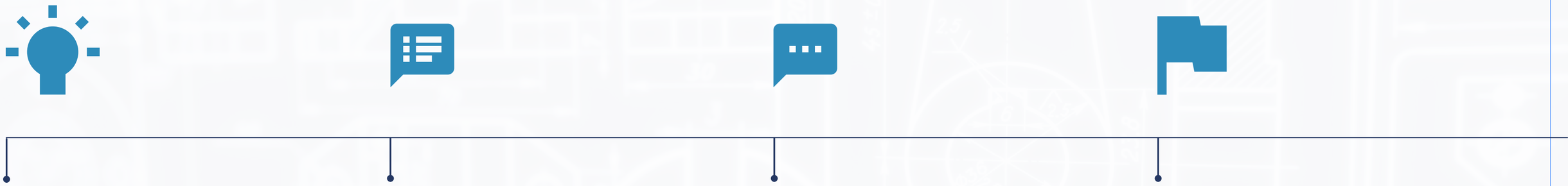
Grad-CAM visualisations enhance transparency, establishing foundations for ***future clinical validation studies.***

Generalizability

The method can be reused for other projects with imbalanced data (e.g., rare diseases in X-rays).

Timeline & Conclusion

Timeline Snapshot



Nov 2025

Data Preparation & Model Setup

Dec 2025

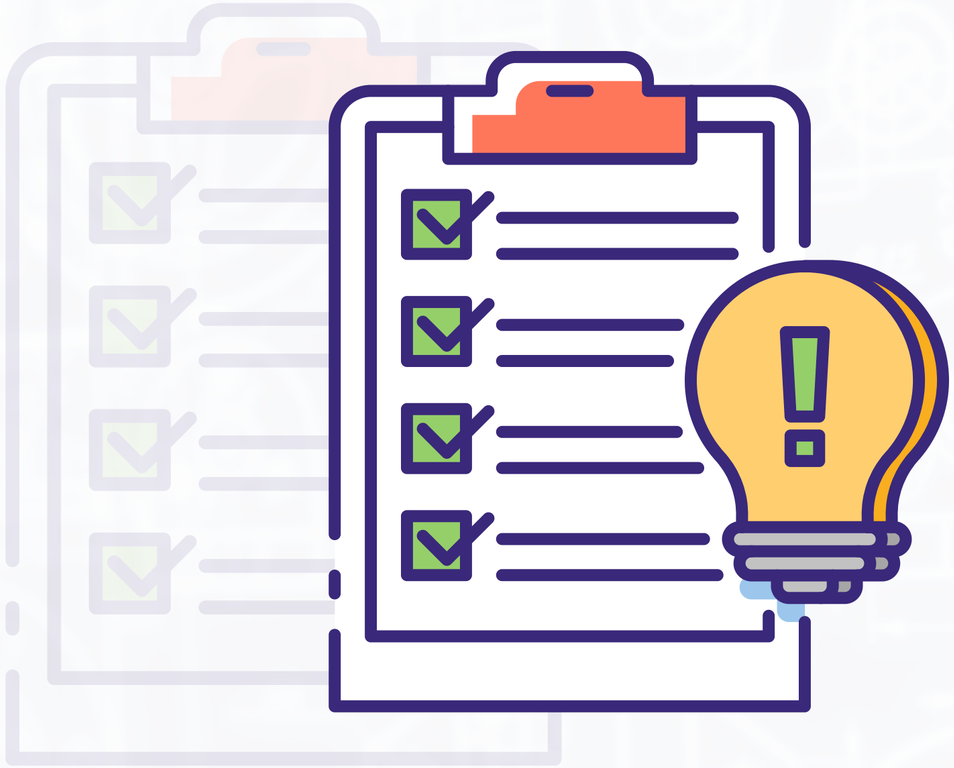
Training & Hyperparameter Tuning

Jan 2026

Evaluation, Error Analysis, Grad-CAM

Feb 2026

Final Thesis Writing & Submission



Conclusion

The proposed method effectively deals with class imbalance by combining **class-specific augmentation, focal loss,** and **attention mechanisms.** This approach improves the model's sensitivity and balance overall.

Limitation

This is designed for **experimental validation and academic investigation** only.

Thank you

For attention